

Ex.No:6	
Date:	

AIM:

ALGORITHM:

CODE:

```
import java.util.*;
```

```
import java.security.*;
```

```
class Main {
```

```
    static long power(long a, long b, long m) {
```

```
        long r = 1;
```

```
        for (int i = 0; i < b; i++)
```

```
        r = (r * a) % m;

    return r;

}
```

```
static long inverse(long a, long m) {

    for (long i = 1; i < m; i++)

        if ((a * i) % m == 1)

            return i;

    return 0;

}
```

```
static long hash(String msg, long q) throws Exception {

    MessageDigest md = MessageDigest.getInstance("SHA-256");

    byte[] h = md.digest(msg.getBytes());

    long value = 0;

    for (byte b : h)

        value = (value * 256 + (b & 255)) % q;

    return value;

}
```

```
public static void main(String[] args) throws Exception {

    Scanner sc = new Scanner(System.in);

    long p = 23, q = 11, g = 4;

    long privateKey = 3;
```

```
long publicKey = power(g, privateKey, p);

long k = 7;

System.out.print("Enter the message: ");

String message = sc.nextLine();

long h = hash(message, q);

// Signature Generation

long r = power(g, k, p) % q;

long s = (inverse(k, q) * (h + privateKey * r)) % q;

System.out.println("\n--- DSA Parameters ---");

System.out.println("p = " + p);

System.out.println("q = " + q);

System.out.println("g = " + g);

System.out.println("\n--- Keys ---");

System.out.println("Private Key = " + privateKey);

System.out.println("Public Key = " + publicKey);

System.out.println("\n--- Signature ---");

System.out.println("Hash = " + h);

System.out.println("r = " + r);

System.out.println("s = " + s);

System.out.println("Signature = (" + r + ", " + s + ")");

// Verification
```

```
long w = inverse(s, q);
```

```
long u1 = (h * w) % q;
```

```
long u2 = (r * w) % q;
```

```
long v = (power(g, u1, p) * power(publicKey, u2, p) % p) % q;
```

```
System.out.println("\n--- Verification ---");
```

```
System.out.println("u1 = " + u1);
```

```
System.out.println("u2 = " + u2);
```

```
System.out.println("v = " + v);
```

```
if (v == r)
```

```
    System.out.println("Signature Status: VALID");
```

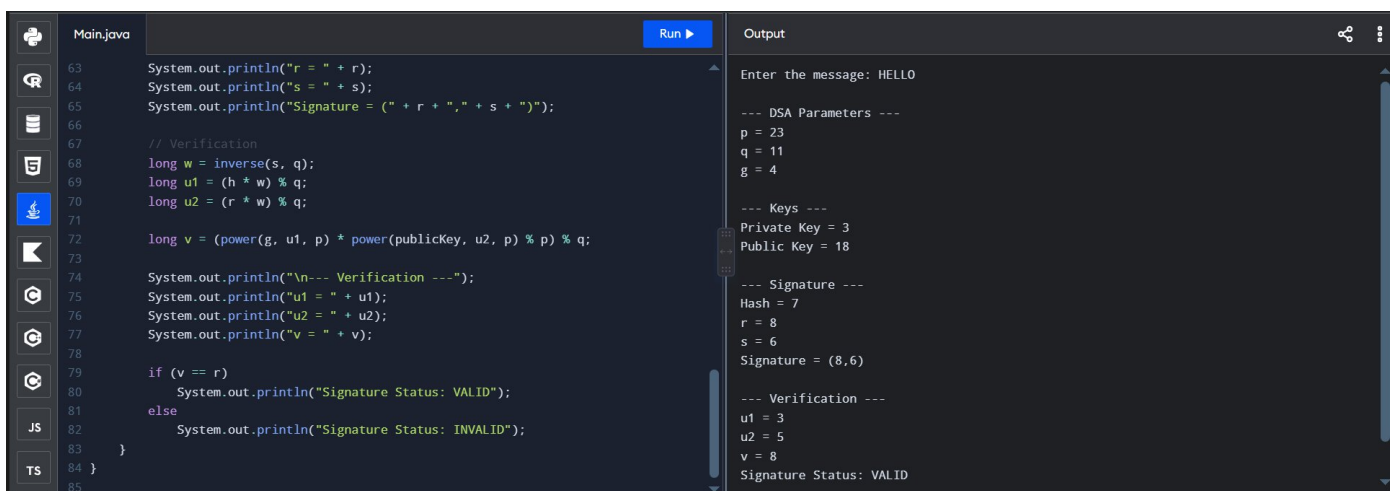
```
else
```

```
    System.out.println("Signature Status: INVALID");
```

```
}
```

```
}
```

Output:



The screenshot shows an IDE with a Java file named 'Main.java' and an 'Output' window. The code in 'Main.java' performs DSA verification. The 'Output' window shows the following text:

```
Enter the message: HELLO  
  
--- DSA Parameters ---  
p = 23  
q = 11  
g = 4  
  
--- Keys ---  
Private Key = 3  
Public Key = 18  
  
--- Signature ---  
Hash = 7  
r = 8  
s = 6  
Signature = (8,6)  
  
--- Verification ---  
u1 = 3  
u2 = 5  
v = 8  
Signature Status: VALID
```

Result: